

ECG_Activity_Recognition_Sadiq_Vaheeb.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[]

```
ecg = nk.ecg_simulate(
    duration=duration,
    sampling_rate=sampling_rate,
    heart_rate=heart_rate,
    method="ecgsyn",
    random_state=42
)

signals[activity] = ecg

print(
    f"{activity.capitalize()} ECG generated: "
    f"{len(ecg)} samples"
)
```

Rest ECG generated: 2500 samples
Walk ECG generated: 2500 samples
Run ECG generated: 2500 samples

[]

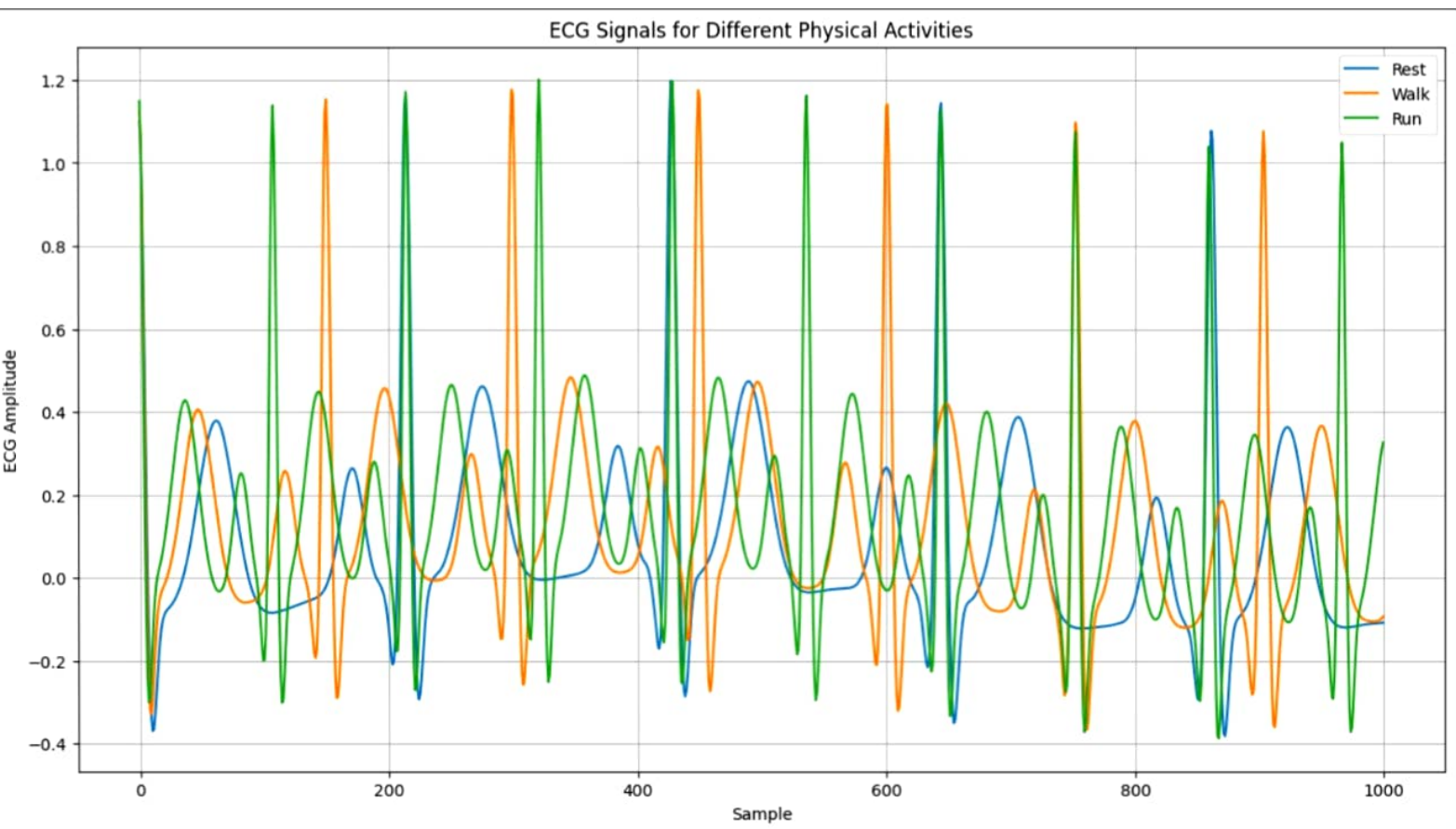
```
plt.figure(figsize=(15, 8))

for activity, ecg in signals.items():

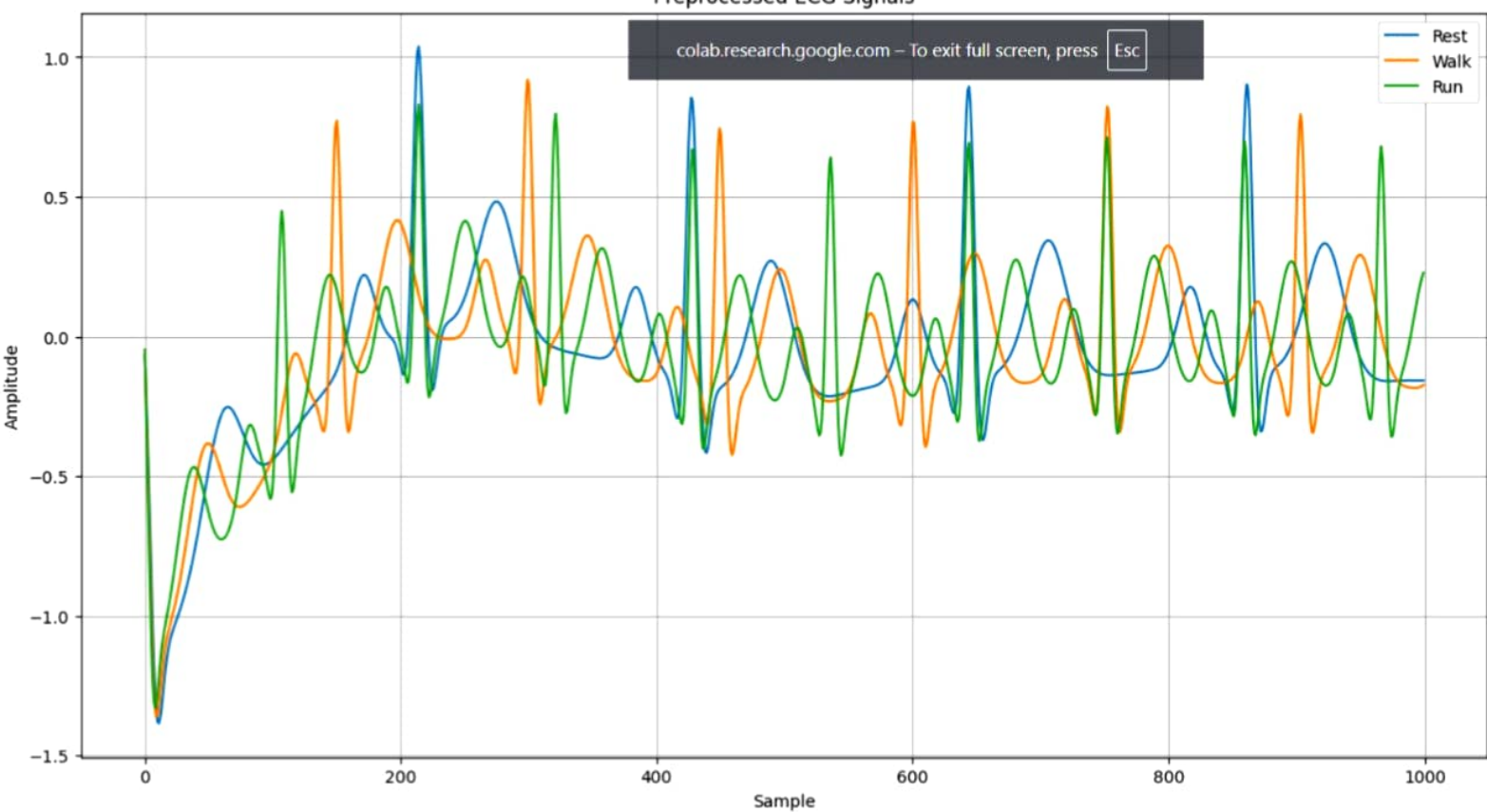
    plt.plot(
        ...
```

Variables Terminal

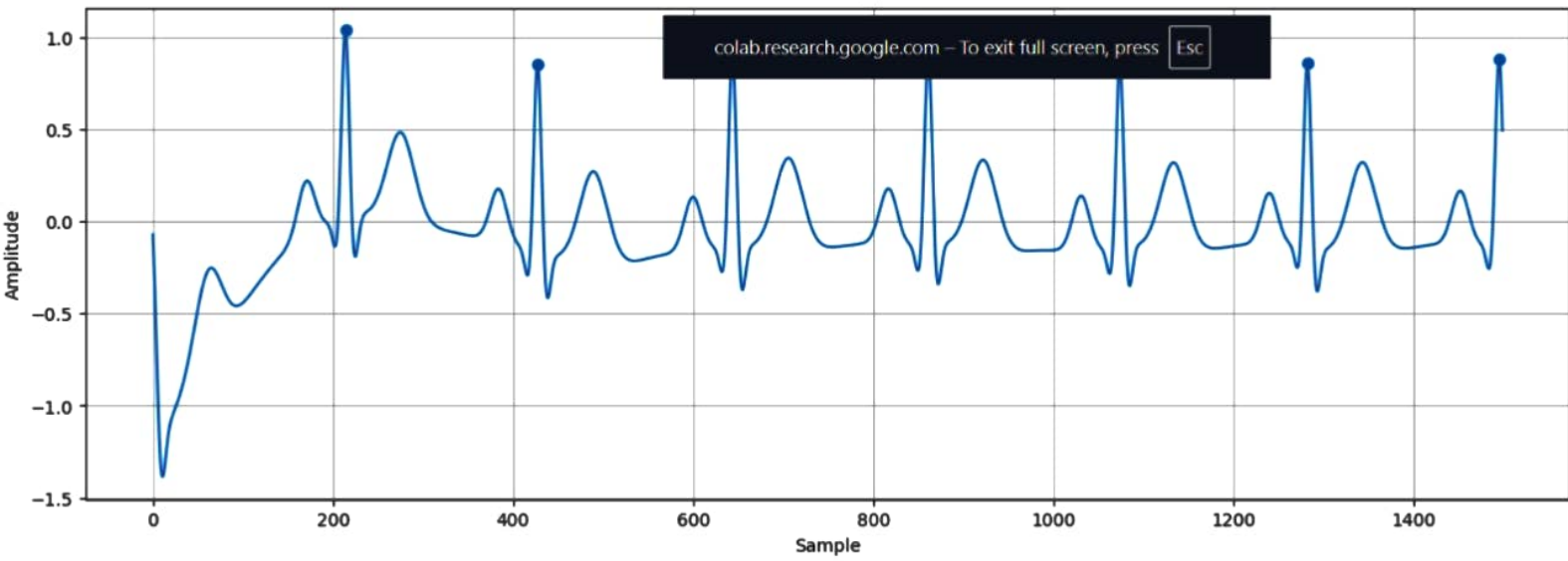
29°C Partly sunny Search 17:44 04-10-2026

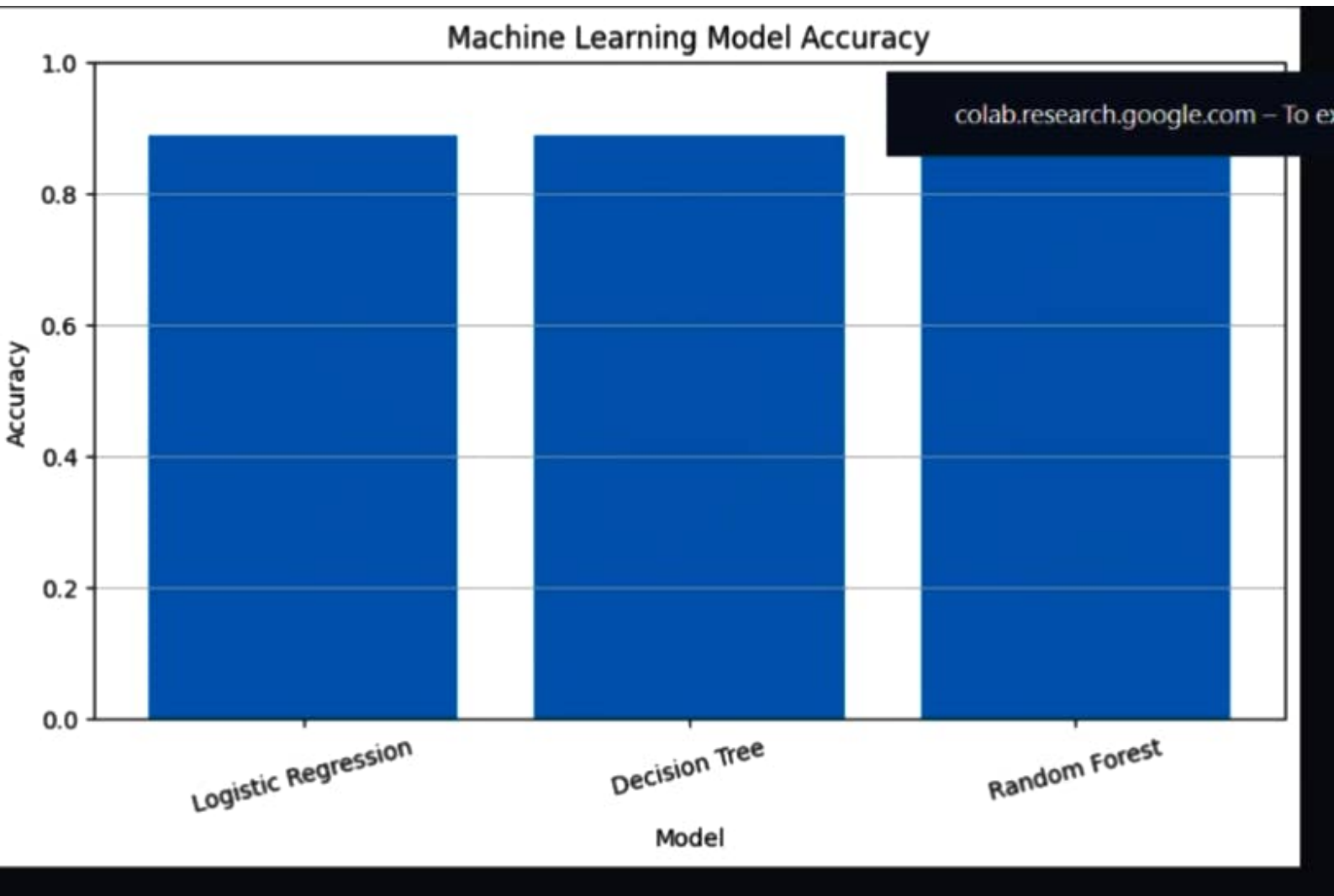


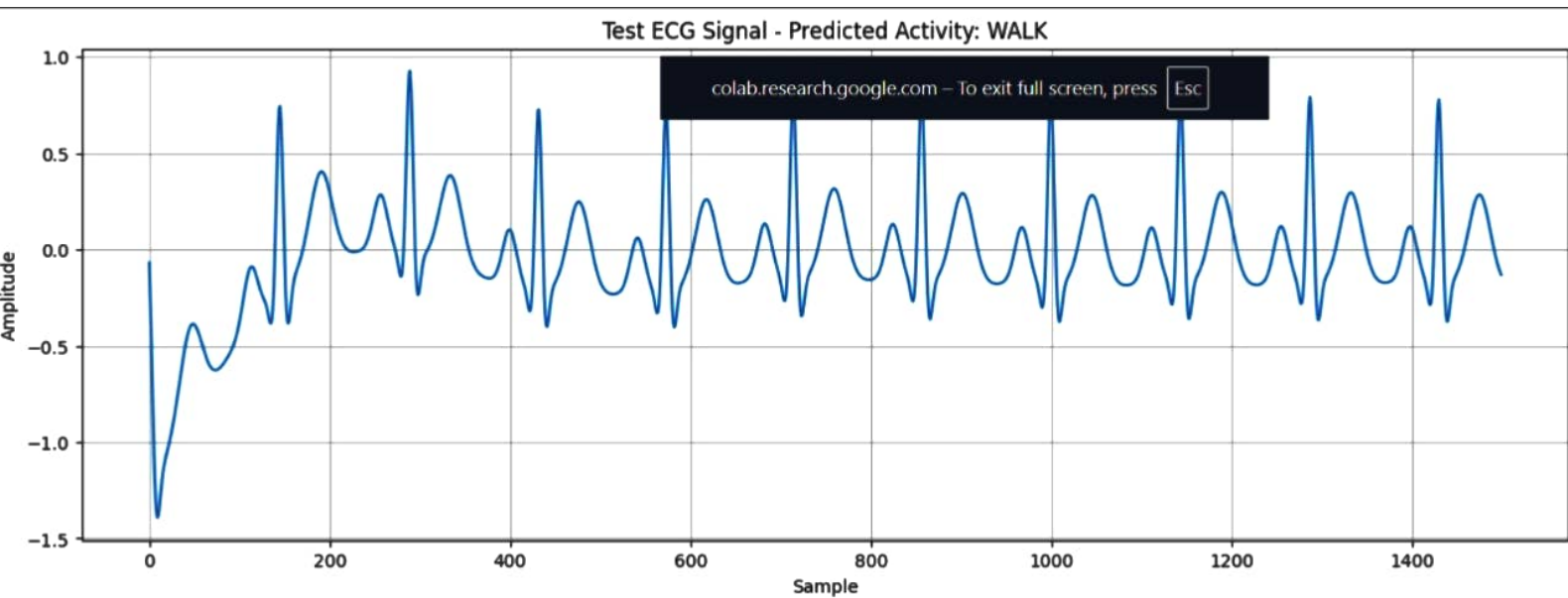
Preprocessed ECG Signals



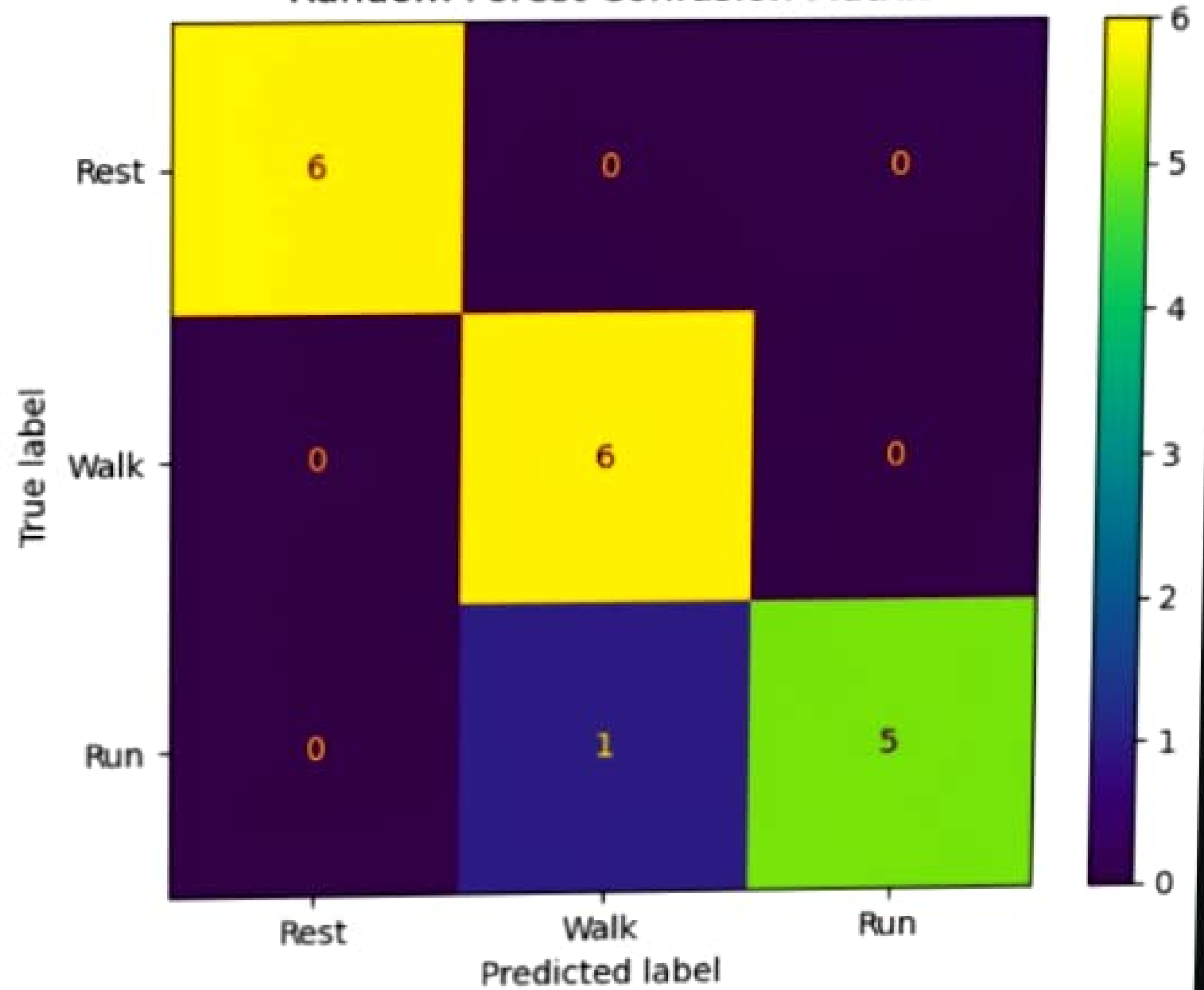
R-Peak Detection - Rest ECG







Random Forest Confusion Matrix



	precision	recall	f1-score	support
rest	1.00	1.00	1.00	6
run	1.00	0.83	0.91	6
walk	0.86	1.00	0.92	6
accuracy			0.94	18
macro avg	0.95	0.94	0.94	18
weighted avg	0.95	0.94	0.94	18